

DATA 201 Group Project:

SQL Analysis & Analytical Queries

Question #1: Room Type Distribution and Average Price Across Los Angeles, San Diego, and San Francisco

SQL Query:

```
SELECT
  c.city_name,
  l.room_type,
  COUNT(l.listing_ID) AS listing_count,
  ROUND(AVG(l.price), 2) AS avg_price,
  ROUND(100.0 * COUNT(l.listing_ID) / SUM(COUNT(l.listing_ID)) OVER (PARTITION BY c.city_name), 2) AS pct_of_city_listings
FROM Listing l
JOIN Neighbourhood n ON l.neighbourhood_id = n.neighbourhood_id
JOIN City c ON n.city_ID = c.city_ID
WHERE l.price BETWEEN 20 AND 1000
GROUP BY c.city_name, l.room_type
ORDER BY c.city_name, listing_count DESC;
```

Result Grid:

city_name	room_type	listing_count	avg_price	pct_of_city_listings
Los Angeles	Entire home/apt	30512	312.45	71.48
Los Angeles	Private room	11384	124.60	26.67
Los Angeles	Hotel room	524	285.12	1.23
Los Angeles	Shared room	268	52.80	0.63
San Diego	Entire home/apt	10742	394.18	82.80
San Diego	Private room	2128	118.35	16.40
San Diego	Hotel room	62	310.50	0.48
San Diego	Shared room	41	49.12	0.32
San Francisco	Entire home/apt	4498	342.15	62.11
San Francisco	Private room	2510	138.90	34.66
San Francisco	Hotel room	182	245.70	2.51
San Francisco	Shared room	52	61.40	0.72

12 row(s) returned | Execution Time: 0.065 sec / 0.0028 sec

Analytical Commentary:

This query examines the room type composition and pricing structure within each market using an analytic window function. It directly explains why San Diego had the highest overall average price in earlier queries: over 82% of its inventory consists of Entire home/apt rentals, compared to just 62% in San Francisco. San Francisco has a significantly higher proportion of private rooms (34.7%), driven by high-density urban housing and strict municipal short-term rental limits. Filtering out placeholder rates under \$20 and above \$1,000 ensures the averages reflect genuine market consumer rates.

Question #2: Top 3 Most Expensive Neighborhoods in Each City (with at least 50 listings)

SQL Query:

```
WITH RankedNeighbourhoods AS (  
  SELECT  
    c.city_name,  
    n.neighbourhood_name,  
    COUNT(l.listing_ID) AS total_listings,  
    ROUND(AVG(l.price), 2) AS avg_price,  
    DENSE_RANK() OVER (  
      PARTITION BY c.city_name  
      ORDER BY AVG(l.price) DESC  
    ) AS price_rank  
  FROM Listing l  
  JOIN Neighbourhood n ON l.neighbourhood_id = n.neighbourhood_id  
  JOIN City c ON n.city_ID = c.city_ID  
  WHERE l.price BETWEEN 20 AND 1500  
  GROUP BY c.city_name, n.neighbourhood_id, n.neighbourhood_name  
  HAVING COUNT(l.listing_ID) >= 50  
)  
SELECT  
  city_name,  
  neighbourhood_name,  
  total_listings,  
  avg_price,  
  price_rank  
FROM RankedNeighbourhoods  
WHERE price_rank <= 3  
ORDER BY city_name, price_rank ASC;
```

Result Grid:

city_name	neighbourhood_name	total_listings	avg_price	price_rank
Los Angeles	Bel-Air	84	894.20	1
Los Angeles	Beverly Crest	246	812.55	2
Los Angeles	Hollywood Hills West	812	625.40	3
San Diego	La Jolla	1180	520.15	1
San Diego	Mission Bay	1420	485.60	2
San Diego	Pacific Beach	1645	398.22	3
San Francisco	Presidio Heights	54	512.80	1
San Francisco	Sea Cliff	61	485.10	2
San Francisco	Pacific Heights	218	440.35	3

9 row(s) returned | Execution Time: 0.071 sec / 0.0031 sec

Analytical Commentary:

By utilizing a Common Table Expression (CTE) combined with DENSE_RANK(), this query isolates the premier micro-markets in each city. A minimum threshold of 50 listings is enforced via the HAVING clause to eliminate statistical skew caused by boutique luxury areas with only one or two listings. The results clearly reflect geographic premiums: luxury foothill estates in Los Angeles (Bel-Air, Beverly Crest), coastal resort zones in San Diego (La Jolla, Mission Bay), and affluent hilltop enclaves in San Francisco (Presidio Heights, Pacific Heights).

Question #3: Impact of Minimum Stay Requirements on Pricing and Reviews

SQL Query:

```
SELECT
  CASE
    WHEN l.minimum_nights = 1 THEN '1 Night (Flexible)'
    WHEN l.minimum_nights BETWEEN 2 AND 6 THEN '2-6 Nights (Short-Term)'
    WHEN l.minimum_nights BETWEEN 7 AND 29 THEN '7-29 Nights (Weekly)'
    ELSE '30+ Nights (Long-Term / Corporate)'
  END AS stay_category,
  COUNT(l.listing_ID) AS total_listings,
  ROUND(AVG(l.price), 2) AS avg_nightly_price,
  ROUND(AVG(l.availability_365), 1) AS avg_annual_availability,
  ROUND(AVG(l.number_of_reviews), 1) AS avg_reviews_per_listing
FROM Listing l
WHERE l.price BETWEEN 20 AND 1000
GROUP BY stay_category
ORDER BY avg_nightly_price DESC;
```

Result Grid:

stay_category	total_listings	avg_nightly_price	avg_annual_availability	avg_reviews_per_listing
2-6 Nights (Short-Term)	31482	328.75	184.2	48.6
1 Night (Flexible)	5618	264.10	168.4	54.2
7-29 Nights (Weekly)	3840	238.45	210.6	28.1
30+ Nights (Long-Term / Corporate)	22114	198.30	245.8	14.5

4 row(s) returned | Execution Time: 0.058 sec / 0.0025 sec

Analytical Commentary:

This query categorizes listings into distinct operational business models based on minimum night thresholds. The 30+ night segment comprises a substantial portion of the database (over 22,000 listings), reflecting regulatory adjustments where hosts switch to intermediate-term leases to circumvent municipal short-term rental restrictions. These 30+ day properties show significantly lower average nightly rates (\$198.30) and far lower review accumulation (14.5 reviews on average) compared to high-turnover weekend vacation rentals.

Question #4: Superhost Prevalence Across Host Portfolio Scales

SQL Query:

```

WITH HostPortfolio AS (
    SELECT
        h.host_ID,
        h.host_is_superhost,
        COUNT(l.listing_ID) AS total_properties
    FROM Host h
    JOIN Listing l ON h.host_ID = l.host_ID
    WHERE h.host_is_superhost IS NOT NULL
    GROUP BY h.host_ID, h.host_is_superhost
)
SELECT
    CASE
        WHEN total_properties = 1 THEN 'Single-Property Host (1)'
        WHEN total_properties BETWEEN 2 AND 5 THEN 'Small Operator (2-5)'
        WHEN total_properties BETWEEN 6 AND 20 THEN 'Medium Operator (6-20)'
        ELSE 'Commercial Megahost (>20)'
    END AS host_scale,
    COUNT(host_ID) AS host_count,
    SUM(CASE WHEN host_is_superhost = 't' THEN 1 ELSE 0 END) AS superhost_count,
    ROUND(100.0 * SUM(CASE WHEN host_is_superhost = 't' THEN 1 ELSE 0 END) / COUNT(host_ID), 1) AS superhost_percentage
FROM HostPortfolio
GROUP BY host_scale
ORDER BY superhost_percentage DESC;

```

Result Grid:

host_scale	host_count	superhost_count	superhost_percentage
Small Operator (2-5)	4892	2495	51.0
Single-Property Host (1)	24115	10852	45.0
Medium Operator (6-20)	984	364	37.0
Commercial Megahost (>20)	347	83	23.9

4 row(s) returned | Execution Time: 0.082 sec / 0.0032 sec

Analytical Commentary:

By aggregating hosts based on their actual portfolio scale, this query investigates whether commercialization impacts hospitality quality indicators. Small operators managing 2-5 units boast the highest Superhost rate at 51.0%, demonstrating professionalized management while still maintaining personalized attention. In contrast, large commercial megahosts (>20 listings, such as corporate property managers) show a steep decline to just 23.9% Superhost status, likely caused by standardized operations, automated messaging, and slower dispute turnaround times.